Sai Tejas S

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Introduction

Aspiring robotics engineer with a solid foundation in robotics, eager to contribute to the manufacturing, aerospace and aviation industries. Currently completing my engineering degree and gaining hands-on experience at a startup. A multilingual communicator, I am naturally curious and passionate about exploring cutting-edge technologies. I stay updated on local and global developments to broaden my perspective and excel in roles that involve leadership, learning, and teamwork.

Education

Sathyabama Institute of Science and Technology, 2021 – 2025

BE CSE specialization in AI and Robotics

Experience

Robot Software Developer Intern, Kelvin6k Pvt Ltd., Chennai Dec 2024 – Present

Kelvin6k is a robotics-based 3D printing construction startup. As a full-time intern, I am gaining valuable hands-on experience in building a robot from scratch, including structural improvements, wiring, motors, drivers, and software development.

Lead of Finance and Marketing, TEDxSIST – Sathyabama IST, Chennai Nov 2024 – Present

Led a team of three in securing sponsorships, building the event budget, managing communication, and developing marketing and social media strategies. Also planned merchandise for the attendees to enhance their event experience.

Event Organizer, Sathyabama Center for Advanced Studies (SCAS) – Sathyabama IST, Chennai Nov 2024 – present

- Experienced in event organization, combining skills in planning, execution, and team coordination to deliver impactful and engaging events.
- Engaged with industry leaders and top academicians to showcase our projects and labs. Organized visits from prominent industries, NAAC, UGC, government officials, celebrities, and accreditation board members.

Volunteer, The American Center, Chennai Jan-2024

Volunteered for The American Center, US Consulate stall at The Chennai Book Fair, where I engaged with visitors, explaining the purpose and initiatives of the center. Assisted in registering memberships while organizing interactive games and activities for kids and attendees.

Publications (On Process)

Autonomous Robot for Crop Cultivation and Harvesting
Journal of Biosystems Engineering
Vismaya V Nair, ***Sai Tejas S***, Dr S Vigneshwari

Projects

Constructed a rover using the MCU 8266 node enabled by Wi-Fi.

- Collaborated with a team of four to build a small-scale rover designed to autonomously measure soil moisture. I was responsible for wiring and assembly while also supporting my teammates. Our project secured 2nd place.

Factory automation using Robo DK

- I completed a freelance project for a master's student at the University of Birmingham, where I was tasked with planning the assembly and disassembly automation of an actuator using RoboDK software. This project was part of the student's final-year work, and my role involved creating an efficient automated system for the assembly and disassembly of the product. (project video link)

Programming Mitsubishi RV-CRL industrial robot to perform various operations

- I independently learned the operation and programming of the robot with the support of customer service. I performed various tasks such as letter and shape tracing, pick and place, and palletization using the proprietary software RT Tool Box 3, which employs their programming language, MELFA BASIC VI. Additionally, I successfully integrated the robot with open-source ROS-based software, RoboDK.

Programming a Humanoid robot NAO to understand its capabilities

- I learned to program the humanoid robot NAO by SoftBank Robotics to explore its capabilities, including object detection, facial recognition, human interaction, and motion control, using their software, Choregraphe.

Built a RC fixed wing plane using cardboard.

- I built a simple remote-controlled fixed-wing plane using cardboard, applying my understanding of the science behind flight. The plane flew successfully, and I controlled it using a transmitter.

Designing a gripper for harvesting vegetables.

- For my final-year project, my teammate and I are developing an autonomous solution for crop harvesting. I designed a soft gripper, using TPU material to 3D print the gripper fingers. In addition, it features a small vacuum gripper for improved handling. In our initial tests, the vegetables were successfully plucked without damage. While I focus on the gripper design, my teammate is working on the software for crop ripeness detection.

Simple Webpages using HTML, CSS and JS.

- I designed several webpages based on my interest's using HTML, CSS, and JavaScript. For some pages, I used ChatGPT to enhance the user experience. You can check them out on my GitHub or website, as mentioned.
- GitHub: https://github.com/saitejass/

Certificates

NPTEL

- Inspection and Quality Control in Manufacturing-IIT Roorkee
- Automation in Manufacturing (NPTEL)-IIT Guwahati

Media Literacy by Learn to Discern-India Academy

- Learn-to-Discern-India Academy is funded by the US Embassy New Delhi and implemented by IREX, aiming to empower the youth leaders of tomorrow by building their resilience to manipulative information and enhancing their critical thinking skills.
- Additional certifications, such as **Youth Trainer** and **Sustained Engagement**, for effective outreach and maintaining consistent involvement.
- Completed projects as part of the program requirements. Developed a board game independently to understand various concepts of media literacy and contributed to drafting an MUN document.

University

- Reverse Engineering using 3D Scanners and 3D Printers
- RC Aircraft Design Training by Feynman Aerospace
- "ROBOTICS" conducted by KODACY
- Workshop on Python
- Drone Design with 3D experience by TANCAM

Seminars and Talks

- Robot programming of Mitsubishi RV-8CRL-DS15 Industrial Robot using RT Toolbox3 software powered by MELFA-BASIC VI programming language.
- Basic projects using RoboDK
- Seminar on Media literacy for professors and students
- Introduction to robotics, Drive Systems etc.

Languages

- English
- Telugu
- Tamil
- Kannada
- Hindi

Technical Skills

- Robot programming (Melfa Basic -VI)
- Robot Simulation (RT toolbox, ABB Robot Studio, RoboDK).
- CAD (CATIA, SolidWorks)
- Python, C++ (Intermediate)
- Additive Manufacturing
- Web Designing (Intermediate)
- ROS (Learning)

Soft Skills

- Team work
- Leadership
- Current Affairs
- Problem Solving
- Social Media
- Event Organizing

Hobbies

- Reading Newspapers/articles
- Seminars and Outreach programs
- Songs and Movies
- Knowing and getting experience with new Technologies
- Knowing more about aeronautics and aerospace
- Teardown of electronic devices
- MUN and structured debate